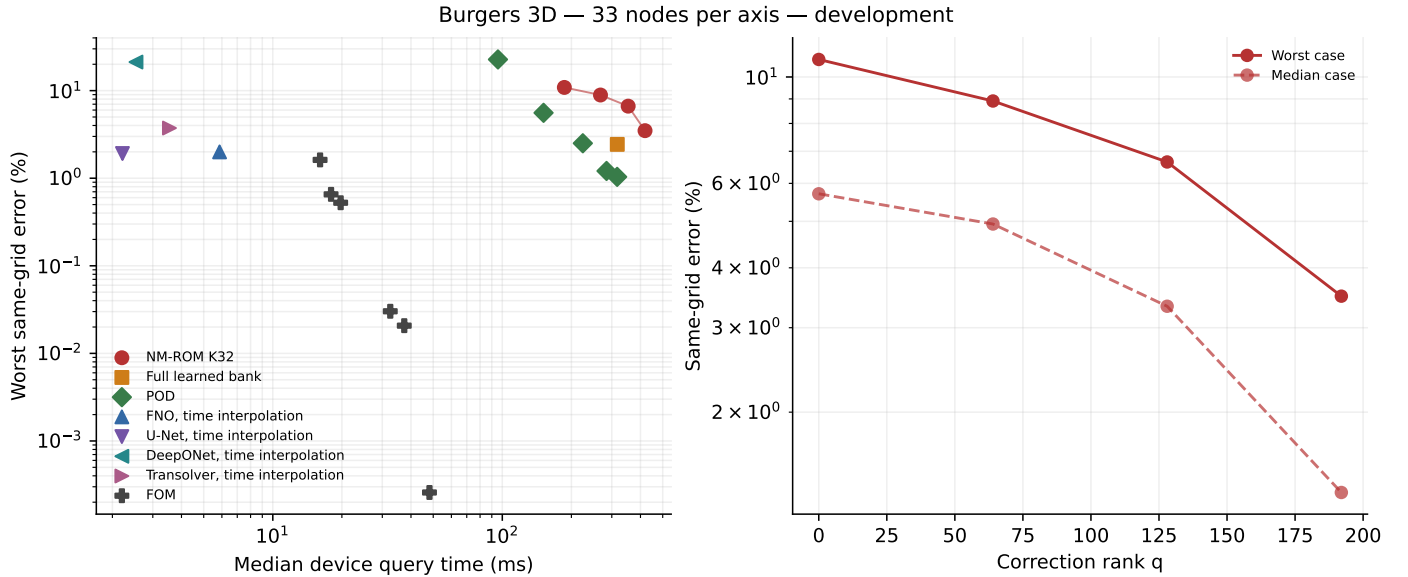


3D comparison panels — provisional development results

All values are generated from retained, audited invocation records. Final evaluation remains pending.

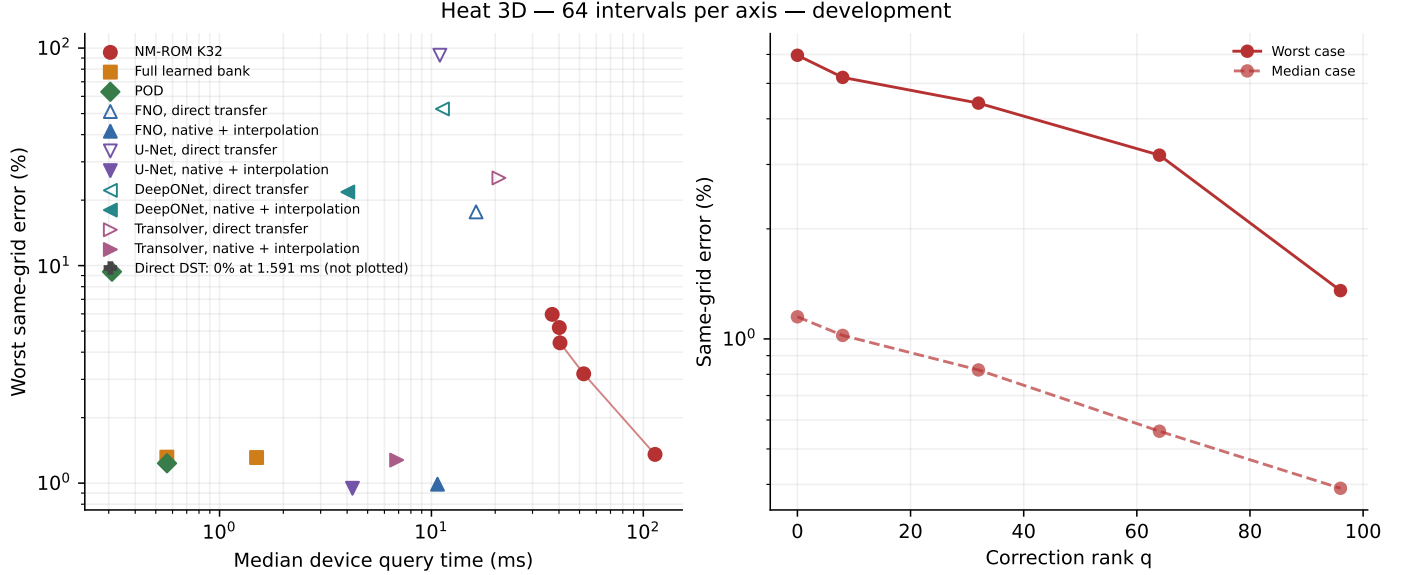
Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K32, q0	5.7071	10.8780	186.364	230.973	0/0	0/24
NM-ROM K32, q64	4.9359	8.9108	267.817	325.616	0/0	0/24
NM-ROM K32, q128	3.3258	6.6491	353.542	428.614	0/0	0/24
NM-ROM K32, q192	1.3600	3.4908	418.712	620.808	0/0	0/24
Full learned bank R256	0.8437	2.4345	316.964	317.629	0/0	0/24
POD32	9.5224	22.6947	95.738	96.211	0/0	0/24
POD96	1.9855	5.5785	151.273	152.382	0/0	0/24
POD160	0.7675	2.5057	224.346	225.287	0/0	0/24
POD224	0.3988	1.2100	284.540	285.347	0/0	0/24
POD256	0.3292	1.0363	316.567	317.512	0/0	0/24
FNO, time interpolation	1.6413	1.9979	5.862	6.687	0/0	0/24
U-Net, time interpolation	1.6239	1.9129	2.212	2.633	0/0	0/24
DeepONet, time interpolation	7.9145	21.1988	2.534	3.009	0/0	0/24
Transolver, time interpolation	1.8700	3.7391	3.545	3.888	0/0	0/24
FOM n1e-02, l5e-01	1.2739	1.6185	16.068	16.725	0/0	0/24
FOM n1e-02, l5e-03	0.0382	0.6543	17.940	20.571	0/0	0/24
FOM n1e-03, l5e-01	0.4692	0.5248	19.753	21.495	0/0	0/24
FOM n1e-04, l5e-01	0.0267	0.0303	32.494	34.086	0/0	0/24
FOM n1e-04, l1e-06	0.0188	0.0207	37.394	41.505	0/0	0/24
FOM n1e-06, l1e-01	0.0002	0.0003	48.156	50.230	0/0	0/24

Table 1: Burgers 3D, 33 nodes per axis; provisional development comparison. Each method is evaluated on 8 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited development data with all four operator families and the same training membership. Errors use the initial-field norm and exclude time zero. Operator predictions interpolate trained time knots to the common output times. The newer operator schedules are still under evaluation. Physical refinement covers only two cases; this panel supports same-grid comparisons only.



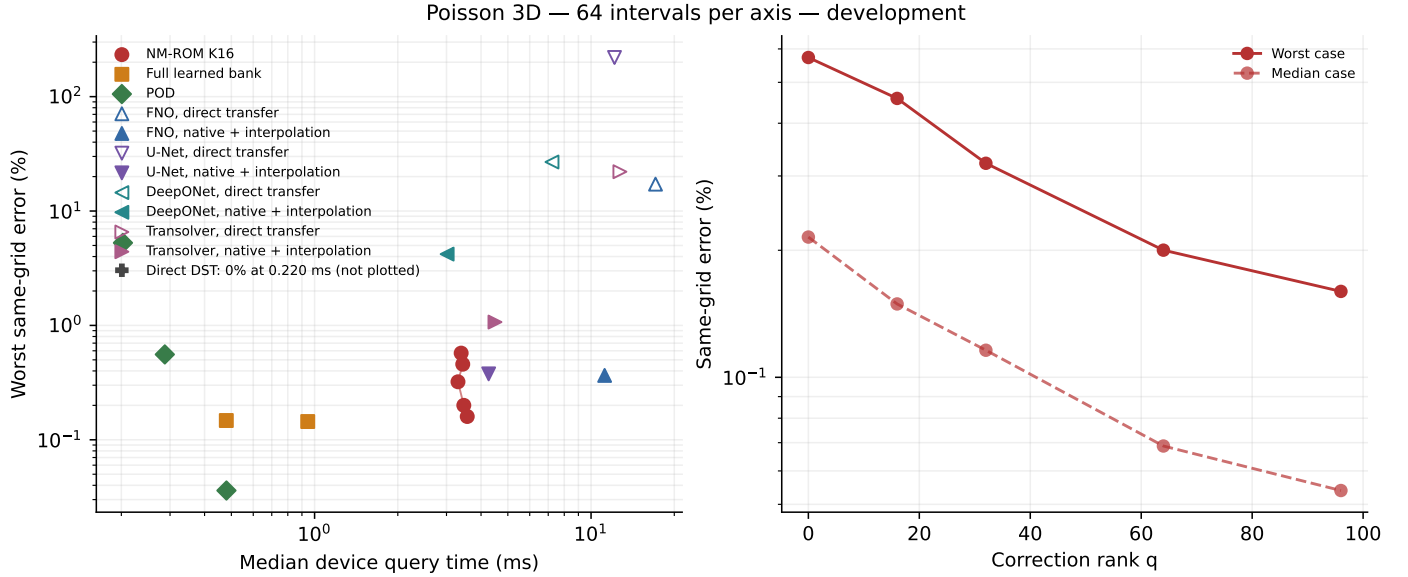
Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K32, q0	1.1494	5.9683	37.143	48.919	0/0	0/48
NM-ROM K32, q8	1.0223	5.1921	40.126	80.283	0/0	5/48
NM-ROM K32, q32	0.8224	4.4151	40.445	56.685	0/0	1/48
NM-ROM K32, q64	0.5591	3.1815	52.322	92.258	0/0	5/48
NM-ROM K32, q96	0.3903	1.3555	113.590	151.651	0/0	1/48
Full bank, exact Galerkin	0.2563	1.3166	0.564	0.766	0/0	1/48
Full bank, exact weak	0.2542	1.3113	1.495	1.694	0/0	0/48
POD32	3.0811	9.3730	0.311	0.647	0/0	9/48
POD128	0.1414	1.2338	0.565	0.616	0/0	0/48
FNO, direct transfer	15.1234	17.6641	16.246	17.031	0/0	0/48
FNO, native + interpolation	0.7669	0.9891	10.681	11.592	0/0	0/48
U-Net, direct transfer	75.7115	92.6756	10.960	11.808	0/0	0/48
U-Net, native + interpolation	0.7512	0.9468	4.240	4.764	0/0	0/48
DeepONet, direct transfer	33.8829	52.5118	11.270	12.017	0/0	0/48
DeepONet, native + interpolation	7.0168	21.7903	4.029	4.545	0/0	0/48
Transolver, direct transfer	19.8616	25.2954	20.828	22.118	0/0	0/48
Transolver, native + interpolation	0.9475	1.2766	6.854	7.952	0/0	0/48
Direct DST	0.0000	0.0000	1.591	1.926	0/0	0/48

Table 2: Heat 3D, 64 intervals per axis; provisional development comparison. Each method is evaluated on 16 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited development data with all four operator families. Errors use the current reference-field norm and exclude time zero. All recorded fits and evolved solves are stationary. The larger head improves the high-correction endpoint but retains a generalization gap; improved initialization and longer operator schedules are under evaluation. Failed direct transfer variants are retained; native-grid interpolation is charged separately. Finer-grid POD is rebuilt offline from the same training members; it is a mesh-adapted classical control, not frozen-weight transfer.



Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K16, q0	0.2149	0.5726	3.390	3.965	0/0	0/48
NM-ROM K16, q16	0.1492	0.4580	3.435	4.269	0/0	0/48
NM-ROM K16, q32	0.1159	0.3215	3.301	4.113	0/0	0/48
NM-ROM K16, q64	0.0687	0.2001	3.465	4.269	0/0	0/48
NM-ROM K16, q96	0.0539	0.1598	3.566	4.219	0/0	0/48
Full bank, weak solve	0.0520	0.1444	0.944	1.049	0/0	0/48
Full bank, Galerkin	0.0530	0.1474	0.479	0.741	0/0	3/48
POD16	1.8304	5.2702	0.203	0.554	0/0	5/48
POD48	0.1484	0.5576	0.287	0.403	0/0	2/48
POD128	0.0066	0.0360	0.480	0.761	0/0	4/48
FNO, direct transfer	16.0680	17.1866	17.107	17.678	0/0	0/48
FNO, native + interpolation	0.2852	0.3656	11.197	11.617	0/0	0/48
U-Net, direct transfer	168.0982	219.8928	12.157	12.513	0/0	0/48
U-Net, native + interpolation	0.2828	0.3769	4.262	4.712	0/0	0/48
DeepONet, direct transfer	16.7244	26.8325	7.221	7.538	0/0	0/48
DeepONet, native + interpolation	2.1141	4.2024	3.016	3.261	0/0	0/48
Transolver, direct transfer	16.9674	22.0653	12.719	13.382	0/0	0/48
Transolver, native + interpolation	0.7201	1.0698	4.493	5.311	0/0	0/48
Direct DST	0.0000	0.0000	0.220	0.501	0/0	6/48

Table 3: Poisson 3D, 64 intervals per axis; provisional development comparison. Each method is evaluated on 16 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited development data with all four operator families. Static weights are placed on the GPU during setup. New operator curves are still improving and longer common schedules are under evaluation. Direct transfer variants and quadrature certificates fail; preserved-domain FNO padding is a pending control. Every measured pair below shares one allocation. Finer-grid POD is rebuilt offline from the same training members; it is a mesh-adapted classical control, not frozen-weight transfer. The saved stopping records are internally consistent, but selected latent states were not retained for an independent weak-gradient audit.



Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K16, q0	28.5319	50.4758	779.231	781.497	0/0	0/24
NM-ROM K16, q16	28.2539	49.9583	810.785	812.390	0/0	0/24
NM-ROM K16, q32	27.8899	49.2381	970.197	973.237	0/0	0/24
NM-ROM K16, q64	27.0931	47.6297	1297.298	1302.637	0/0	0/24
NM-ROM K16, q128	25.2664	45.3348	1865.999	1873.513	0/0	0/24
Full bank, Galerkin R512	13.4503	24.2696	43.878	44.156	0/0	0/24
POD16, weak solve	87.7996	91.9857	25.084	26.989	0/0	0/24
POD32, weak solve	78.1203	85.5263	27.511	28.055	0/0	0/24
POD48, weak solve	67.3545	76.1070	35.081	35.300	0/0	0/24
POD128, Galerkin	34.5481	47.3334	5.609	5.832	0/0	0/24
POD256, Galerkin	24.8410	36.0594	11.294	11.345	0/0	0/24
POD512, Galerkin	15.2837	27.7537	43.895	44.105	0/0	0/24
FOM, dt 0.001	0.0031	0.0051	24.712	25.475	0/0	0/24
FOM, dt 0.002	0.0130	0.0213	12.810	13.163	0/0	0/24
FOM, dt 0.004	0.0524	0.0866	6.886	7.224	0/0	0/24
FOM, dt 0.008	0.2125	0.3598	3.893	4.227	0/0	0/24
FNO, raw	1.3422	1.7664	8.787	10.315	0/0	0/24
FNO, projected	1.0974	1.5121	8.970	12.449	0/0	1/24
U-Net, raw	3.4540	4.6468	1.982	2.112	0/0	0/24
U-Net, projected	3.0414	4.2830	2.094	2.212	0/0	0/24
DeepONet, raw	36.8223	45.3970	4.124	4.362	0/0	0/24
DeepONet, Leray projected	35.8894	44.8450	4.372	4.600	0/0	0/24
Transolver, raw	2.2017	2.9715	3.317	4.436	0/0	1/24
Transolver, Leray projected	1.3773	2.0366	3.398	4.553	0/0	1/24

Table 4: Navier–Stokes 3D, 32 periodic points per axis; provisional development comparison. Each method is evaluated on 8 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Independently audited new paired development measurements with all four operator families. Errors use the initial vector-field norm and exclude time zero. Learned-bank and head accuracy targets fail. Saved latent histories support sampled independent weak-gradient checks. A strict cross-run frozen-field replay gate failed and remains failed; no numerical-equivalence or implementation-speed claim is made. Augmented training and larger-bank tuning remain in progress.

